

***par-1*, a Gene Required for Establishing Polarity in *C. elegans* Embryos, Encodes a Putative Ser/Thr Kinase That Is Asymmetrically Distributed**

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Summary

The first cleavage of *C. elegans* is asymmetric, generating daughter cells with different sizes, cytoplasmic components, and fates. Mutations in the *par-1* gene disrupt this asymmetry. We report here that *par-1* encodes a putative Ser/Thr kinase with similarity to kinases from yeasts and mammals. Two strong alleles have mutations in the kinase domain, suggesting that kinase activity is essential for *par-1* function. PAR-1 protein is localized to the posterior periphery of the zygote and is distributed in a polar fashion preceding the asymmetric divisions of the germline lineage. Because PAR-1 distribution in the germline correlates with the distribution of germline-specific P granules, it is possible that PAR-1 functions in germline development as well as in establishing embryonic polarity.

Introduction

Asymmetric cell division, by which the two daughter cells adopt different fates, plays an important role in the generation of cellular diversity during development. Daughter cells with different fates can arise via extrinsic cell–cell signaling or by intrinsically asymmetric divisions (Horvitz and Herskowitz, 1992). Intrinsically determined asymmetric divisions occur in a number of invertebrates and appear to play crucial roles in establishing embryonic axes or specifying particular differentiated cell fates (Davidson, 1986; Slack, 1991). Most evidence supports the model of cytoplasmic localization as the basis for such intrinsically patterned asymmetric divisions. In cytoplasmic localization, maternally provided “determinants” are unequally partitioned to particular cells through a series of reproducibly oriented cleavage divisions (Davidson, 1986).

The nematode *Caenorhabditis elegans* provides an opportunity to study the mechanisms responsible for intrinsically asymmetric divisions. *C. elegans* embryos have a fixed cell lineage, and a precise cell fate map has been established (Sulston et al., 1983). The asymmetric divisions leading to the formation of the germline play an important role in establishing embryonic organization. Each of these divisions is asymmetric, producing daughter cells with differences in size, cell division timing, cleavage pattern, and ultimate fate (see Figure 1A for a summary of the cell lineage). In addition, germline-specific P granules are localized to one pole of the cell prior to each division and partitioned into the smaller daughter cell (Strome and Wood, 1982, 1983).

Although the mechanisms underlying this series of asymmetric divisions are largely unknown, microfilaments

appear to play a role in at least the first division. Brief pulses of the microfilament-disrupting drug cytochalasin during a critical period of the first cell cycle prevent the posterior localization of the P granules (Strome and Wood, 1983; Hill and Strome, 1988). Cytochalasin pulses during this same period sometimes also lead to symmetric divisions producing daughter cells of equal sizes, similar cell cycle rates, and variable spindle orientations (Hill and Strome, 1988, 1990).

Maternal-effect lethal mutations in the *par* genes produce phenotypes similar to the effects of cytochalasin treatments (Kemphues et al., 1988). In *par* mutant embryos, normally unequal divisions are equal, cleavage spindles are misoriented, P granules are mislocalized, and the blastomere fates are altered. This implies that the *par* genes function in both spindle placement and cytoplasmic localization (Kemphues et al., 1988; Kirby et al., 1990; Morton et al., 1992; Cheng et al., 1995).

The *par-1* gene is required for several aspects of early embryonic polarity (Kemphues et al., 1988). As schematically diagrammed in Figure 1B, *par-1* mutant embryos exhibit equal first cleavage and fail to localize P granules. The 2-cell stage blastomeres divide synchronously, and the cleavage patterns are aberrant for all subsequent stages. The *par-1* embryos arrest as amorphous masses with large numbers of differentiated cells. Most embryos contain excess body wall and pharyngeal muscle cells (normally produced by the MS lineage) and are completely lacking intestine. The putative MS cell fate determinant, SKN-1, normally distributed asymmetrically among early blastomeres, is found in equal amounts in all the cells at the 2- and 4-cell stages in *par-1* mutant embryos (see Figure 1B). The excessive pharyngeal and body wall muscle cells observed in *par-1* mutant embryos may arise as a consequence of the abnormal distribution of SKN-1 (Bowerman et al., 1993).

We demonstrate here that *par-1* encodes a conserved putative Ser/Thr kinase that is asymmetrically localized to the posterior periphery of the 1-cell embryo and differentially distributed, preceding all the asymmetric divisions of the germline lineage. Mutations in the kinase domain eliminate PAR-1 function, suggesting PAR-1 kinase activity is essential for its role in establishing asymmetry.

Results

Molecular Cloning of the *par-1* Gene Identification of the *par-1* Region

As summarized in Figure 2A, *par-1* maps to chromosome V between genetic markers *rol-4* and *unc-76*, an interval of ~3 m.u. Telfer (1991) had placed the *par-1* gene on the *C. elegans* physical map within or near the region covered by the yeast artificial chromosome (YAC) Y39H1. We further mapped *par-1* relative to two closely linked genes, *ogr-1* and *him-5*. The mapping data put *par-1* ~0.075 m.u. (~60 kb) to the right of *him-5* in a region only partially covered by cosmid clones (described in Experimental Pro-

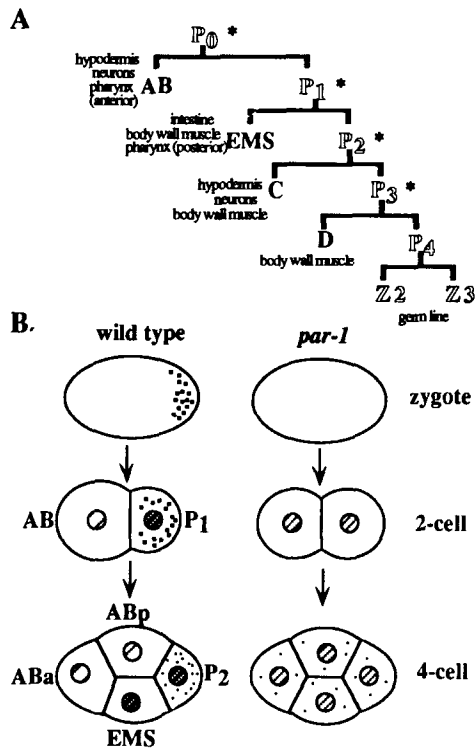


Figure 1. Cell Lineages and Early Divisions in *C. elegans*

(A) Diagram showing a simplified cell lineage and major cell types produced by the founder cells (Sulston et al., 1983). P0 is the zygote. Posterior is to the right. The blastomeres of the P lineage, namely, P0 to Z2 and Z3, are outlined. Those P lineage blastomeres that undergo asymmetric divisions are marked with asterisks. Note that asymmetric division of EMS segregates the ability to produce intestine to the E daughter and the ability to produce pharyngeal and body wall muscles to the MS daughter.

(B) Schematic drawing of 1-, 2-, and 4-cell wild-type (left) and *par-1* mutant (right) embryos, showing the divisions and the distribution of P granules (represented by dots in the cytoplasm) and transcription factor SKN-1 (represented by wavy lines in the nucleus; the extent of shading represents the relative concentration of SKN-1 protein). Posterior is to the right. In wild-type embryos, first and second divisions produce daughter cells differing in size, division timing, cleavage pattern, and cell fates. P granules are localized to the posterior end before the first division; they are segregated into the P1 blastomere by first asymmetric cleavage and further localized to the P2 blastomere (Strome and Wood, 1983). SKN-1 protein is highly concentrated in the P1 nucleus and remains concentrated in the sister blastomeres EMS and P2 derived from P1 (Bowerman et al., 1993). In *par-1* mutant embryos (from homozygous *par-1* mothers), asymmetry in size, division timing, and cleavage pattern is disrupted, and cell fates are dramatically altered. P granules fail to localize. Since the P granules are undetectable in the late 1-cell *par-1* embryos and in the 2-cell *par-1* embryos, they are only shown in the 4-cell embryo. SKN-1 protein is found in equal concentration in all blastomeres at 2-, 4-, and 8-cell stages.

cedures). We were able partially to rescue *par-1* mutants by transformation with the YAC, Y51F3 (see Experimental Procedures). Rescued *par-1* homozygous animals produced about five hatching progeny per worm, but all of these progeny were agametic. This agametic phenotype is a maternal effect, since *par-1* homozygous hermaphrodites derived from *par-1*⁺ mothers produced gametes. We interpret this result to mean that the expression of

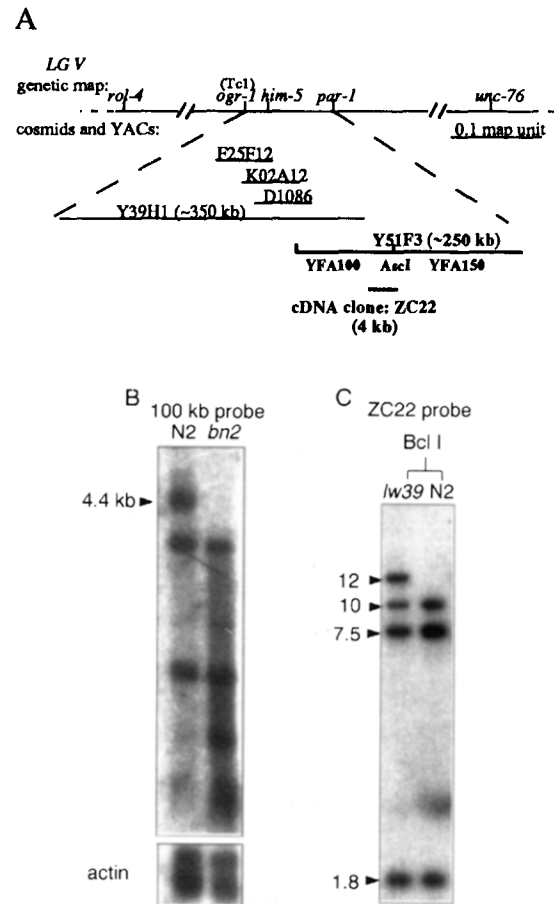


Figure 2. Identification of *par-1*

(A) Placement of *par-1* on the physical map. The diagram shows the genetic map and physical map in the *par-1* region. Genetic mapping puts *par-1* ~0.075 m.u. (estimated ~60 kb) to the right of *him-5* (see Experimental Procedures). The YAC Y51F3 rescued *par-1* mutant embryos in a germline transformation assay. Y51F3 was cut with the restriction enzyme *Ascl*, and two fragments, YFA100 and YFA150, were generated.

(B) Northern blot analysis of maternally enriched transcripts present in the 100 kb fragment of Y51F3. Poly(A)⁺ RNA was isolated from wild-type (N2) and the temperature-sensitive germline-deficient strain *glp-4(bn2)*, resolved by electrophoresis, blotted, and probed with the 100 kb fragments. A single maternally enriched transcript is detected and indicated with an arrowhead. It is ~4.4 kb in length.

(C) Genomic Southern blot analysis of *par-1(lw39)* and wild type (N2). Genomic DNA was isolated from wild-type (N2) and *par-1(lw39)* heterozygous animals, digested with *Bcl*I, resolved by electrophoresis, blotted onto nylon filter, and probed with the cDNA clone ZC22. Three bands, 1.8 kb, 7.5 kb, and 10 kb, were detected in N2; in *par-1(lw39)* heterozygous animals, 7.5 kb and 10 kb bands were found to be at decreased intensity; in addition, a novel band of ~12 kb was detected.

par-1⁺ in the extrachromosomal array is sufficient to rescue some embryos from maternal-effect lethality but not sufficient to prevent a maternal-effect sterility. Consistent with this interpretation, escapers from the maternal-effect lethality of the leaky *par-1* allele *lw7* are also agametic (unpublished data).

Our attempts to identify smaller segments of chromosomal DNA that could rescue *par-1* failed because we were unable to recover clones of DNA that contiguously covered

Table 1. Scoring for Embryonic Lethality in RNA Injections

Molecules Injected	Number of Worms Injected	Percent Embryonic Lethality
ZC22 (<i>par-1</i>) antisense	16	52
ZC22 (<i>par-1</i>) sense	12	54
Clone TS antisense	8	0
Clone pZ1 antisense	8	0
DEPC-dH ₂ O	4	0

ZC22 is the nearly full-length cDNA clone (4 kb) for *par-1*; TS is *twinstar*, a *Drosophila* gene encoding cofilin (K. Gunsalus and M. Goldberg, personal communication); clone pZ1 is a zygotic cDNA clone identified by J. Watts (personal communication). The number of hatched versus unhatched embryos was counted for the first 2 days following the injections (a minimum of 100 embryos from each injected worm were counted).

this region. Therefore, we took an alternative approach to clone the *par-1* gene.

Isolation of a Candidate cDNA and Detection of an Allele-Associated Polymorphism

We cut Y51F3 into 100 kb and 150 kb fragments with the restriction endonuclease *Asc*I. Southern blot analysis using the end of the cosmid D1086 as a probe showed that the 100 kb fragment is proximal to *him-5* (data not shown) and thus likely to contain *par-1*. As *par-1* mutations are strictly maternal (Kemphues et al., 1988), we searched this 100 kb region for the presence of maternally enriched mRNAs that would be candidates for the *par-1* gene. We detected one message of ~4.4 kb that was greatly enriched in wild type relative to the germline-deficient strain, *glp-4(bn2)* (Figure 2B). We isolated cDNAs corresponding to this transcript (see Experimental Procedures). By using the longest cDNA (ZC22) as a probe, we detected an allele-associated restriction fragment length polymorphism (Figure 2C) in the γ -irradiation-induced *par-1* allele, *lw39* (a gift from J. Shaw). Further Southern blot analyses using fragments of the cDNA clone as probes suggest that the lesion in the *lw39* allele is a deletion of ~5 kb in the transcription unit (data not shown). This result indicates that the ZC22 cDNA may be derived from *par-1*.

Phenocopying *par-1* in Wild Type via Antisense RNA Injection

Germline transformation rescue with a genomic clone containing only the gene of interest has been routinely used to confirm cloning of genes in *C. elegans*. However, when we searched for the corresponding genomic clone in a genomic DNA library (provided by H. Browning) or in the YAC Y51F3 subclone libraries we previously constructed, we failed to identify clones that would represent the ZC22 genomic region. Therefore, we decided to determine whether the cDNA we identified corresponds to the *par-1* gene using antisense inhibition. Antisense inhibition has been previously accomplished in transgenic worms carrying antisense DNA constructs (Fire et al., 1991). Because *par-1* was expected to be expressed in the germline, we carried out a more direct inhibition assay by injecting antisense RNA into gonads of wild-type worms (see Experimental Procedures).

The results of these injections are summarized in Table

1. An average of 50% of the embryos from the ZC22 antisense RNA-injected wild-type animals arrested development with *par-1* terminal phenotypes: no morphogenesis, no intestinal cells, and excess of pharynx. Furthermore, most of the 50 or so early embryos we dissected from injected worms exhibited cell division patterns characteristic of *par-1* loss-of-function mutations. For example, many embryos exhibited *par-1*-specific pseudocleavage defects (Kirby et al., 1990), equal first cleavage, and synchronous early divisions (data not shown). Also, consistent with our results from germline transformation, some of the surviving progeny from injected worms were agametic. In control experiments, in which H₂O and antisense RNA made from two unrelated cDNA clones were injected, no effects were observed.

Surprisingly, injection of in vitro synthesized sense RNA from the cDNA ZC22 also induced *par-1* phenotypes at a high frequency among the progeny of injected worms. It is not clear what accounts for this effect. Moreover, the sense effect appears to be restricted to the putative translated region of the RNA while the antisense effect is not. Injection of both sense and antisense RNA from the 5' region (lacking the 3' untranslated region) also gave *par-1* phenocopies, while only the antisense RNA from the 3' untranslated region gave an effect (data not shown). Thus, the antisense and sense effects appear to be separable and probably involve different mechanisms. The basis for the sense effect is under investigation and will not be discussed further. Overall, the specificity of the antisense and sense phenocopies provides strong evidence that the ZC22 cDNA represents the *par-1* transcript. Additional evidence is provided below.

The *par-1* Gene Encodes a Conserved Putative Ser/Thr Kinase

We sequenced the partial cDNA ZC22 and used the rapid amplification of cDNA ends-polymerase chain reaction (RACE-PCR) to identify the 5' end of the transcript (see Experimental Procedures). Four different 5' ends were detected, all of which were *trans*-spliced to the SL1 leader (Krause and Hirsh, 1987; Bektess et al., 1988). The nucleotide sequence of the longest mRNA derived from the RACE-PCR results has been submitted to GenBank and is not reproduced here. A single long open reading frame is detected in the message; the putative start codon is preceded by stop codons, and a *C. elegans* translation initiation consensus sequence, AACAA (M. Perry, personal communication), is found just upstream of this ATG. The predicted PAR-1 protein consists of 1192 amino acids with a molecular mass of ~126 kDa. Protein data base searches reveal that PAR-1 is a putative Ser/Thr kinase highly conserved among different organisms (Figure 3A). PAR-1 shows striking similarity to its mammalian homologs. In the kinase domain, PAR-1 shares ~80% identity with the human Kp78 (217 of 260) (Parsa, 1988) and mouse Emk (220 of 260) (Inglis et al., 1993) and ~46% identity (70% similarity) with yeast SNF1 (128 of 260) (Celenza and Carlson, 1986), the rat AMP-dependent kinase (122 of 260) (Carling et al., 1994), and yeast KIN1/KIN2 (113 of 260) (Levin et al., 1987; Levin and Bishop, 1990). Intriguingly, besides strong homologies in the kinase domain,

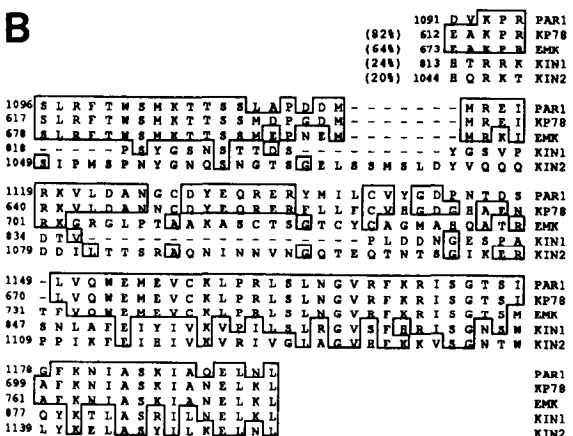
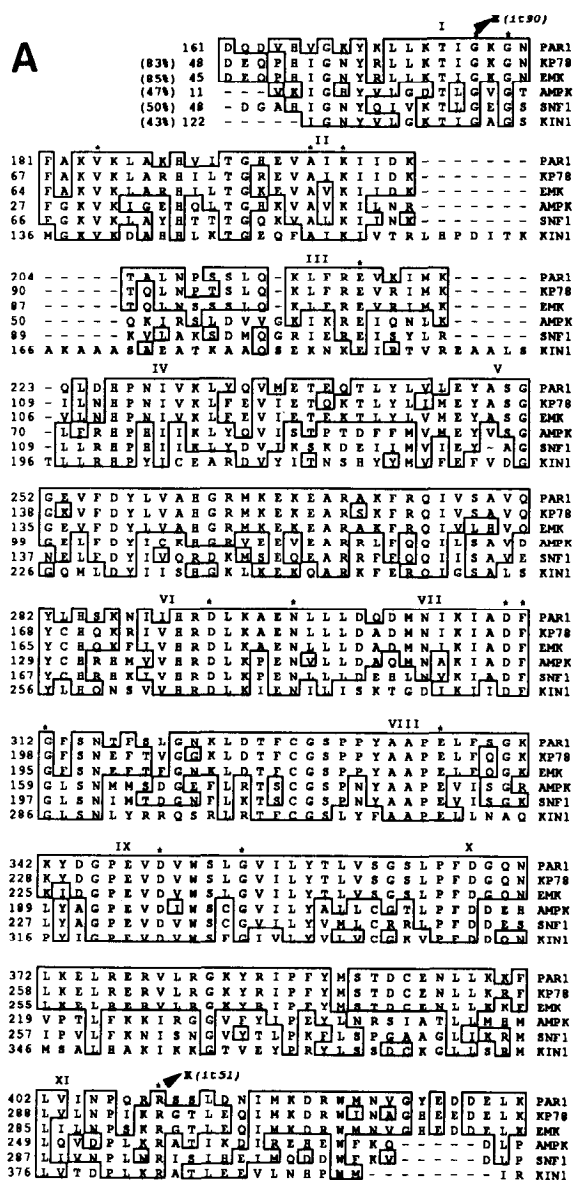


Figure 3. Sequence Alignment of PAR-1 with Members of the SNF1 Protein Kinase Subfamily

(A) Amino acid sequence comparison of the kinase domain from PAR-1 with homologous kinases identified in mammals and yeast: human

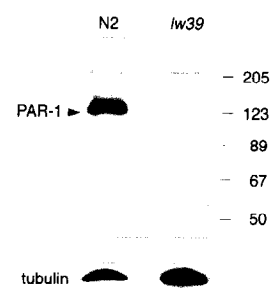


Figure 4. Detection of Endogenous PAR-1 Protein on Western Blots. Extracts from wild-type (N2) and *par-1(lw39)* homozygotes were resolved by SDS-PAGE, blotted onto nitrocellulose filters, and probed with the affinity-purified anti-PAR-1 antibody (see Experimental Procedures). A major band of ~126 kDa (the expected size of the PAR-1 protein) was detected in N2, but not in *par-1(lw39)* homozygotes. The same blot was reprobed with a monoclonal anti- α -tubulin antibody as a loading control. Positions of molecular mass markers are indicated on the right.

an additional conserved domain at the C-termini of the proteins is present in PAR-1, Kp78, Emk, and KIN1/KIN2 (Figure 3B). This domain is highly conserved among PAR-1 and the mammalian kinases, but much diverged in the yeast kinases.

To verify that this gene is indeed *par-1*, we sequenced the kinase domain in 10 mutant alleles and found alterations of invariant kinase residues (Hanks and Quinn, 1991) in two *par-1* mutants. In allele *it90*, an invariant residue Gly of the putative ATP-binding site in the kinase subdomain I is changed to Glu (Figure 3A). In allele *it51*, an invariant residue Arg in subdomain XI is changed to Lys (Figure 3A). Although Arg and Lys are both positively charged, the invariance of this position in all protein kinases that have been sequenced to date suggests that this Arg residue can not be substituted. A similar conserved substitution (Asp to Glu) in an invariant kinase residue has also been reported in a mutant allele of the *ctr1* gene in *Arabidopsis* (Kieber et al., 1993).

PAR-1 Protein Is First Detected in Newly Forming Oocytes

To examine the distribution of PAR-1 in adult hermaphrodites and early embryos, we raised polyclonal antibodies against a portion of the PAR-1 protein (see Experimental Procedures). The antibody specificity was examined on

Kp78 (sequence unpublished but found in GenBank), mouse Emk (Inglish et al., 1993), rat AMP-activated protein kinase (Carling et al., 1994), SNF1 (Celenza and Carlson, 1986), and KIN1 in *S. pombe* (Levin and Bishop, 1990). The alignment was generated using the CLUSTAL method (Higgins and Sharp, 1989) with DNASTAR software. Identical amino acids are boxed. The kinase subdomains as defined by Hanks et al. (1988) are shown. The invariant kinase residues are indicated by asterisks. The percentage of amino acid identity between PAR-1 and other kinases is indicated. The amino acid alterations of invariant residues observed in *it51* and *it90* are indicated by arrowheads.

(B) Alignment of the C-terminal domains in PAR-1, Kp78, Emk, and KIN1/KIN2. Identical residues are boxed, and the percentage of identity is indicated.

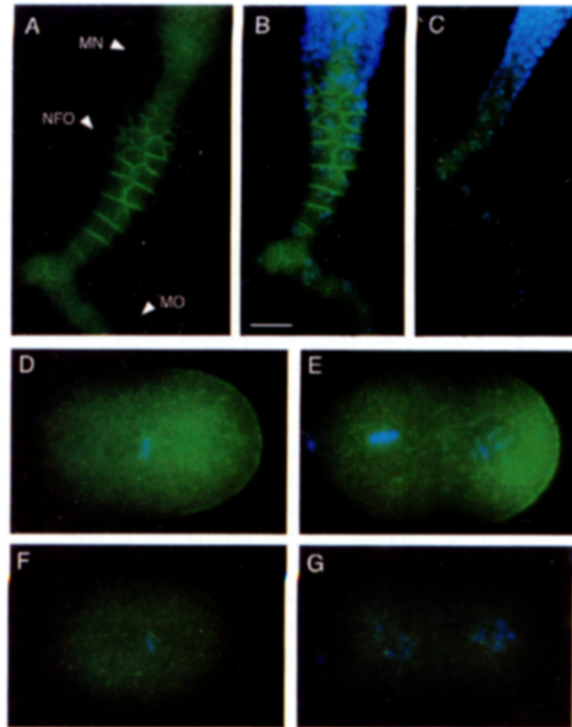


Figure 5. Distribution of PAR-1 in the Gonad and Early Embryo
Immunofluorescence image of gonads and embryos showing the distribution of the anti-PAR-1 antibody (green) and nuclei (blue). (A) Immunofluorescence image of a wild-type gonad showing strong fluorescence at the periphery of newly forming oocytes. Although we detect signal around the periphery of these young oocytes, the outer membrane signal is faint and is not apparent in this reproduction. Abbreviations: MN, meiotic nuclei; NFO, newly forming oocytes; MO, mature oocytes. (B) Double exposure of the same gonad shown in (A), showing the localization of anti-PAR-1 relative to nuclei. (C) Double exposure showing absence of peripheral staining in *par-1(it86)* mutant gonad. (D–E) Wild-type 1- and 2-cell embryos, respectively, showing strong peripheral fluorescence in the posterior. (F–G) Mutant 1- and 2-cell *par-1(lw39)* embryos showing the absence of posterior peripheral staining and the reduction of cytoplasmic staining. The scale bar in (A)–(C) is 32 μ m; in (D)–(G), 8 μ m.

Western blots (Figure 4). Using this antibody, we have stained worms and isolated gonads and early embryos of wild-type and *par-1* mutants (see Experimental Procedures). The patterns we describe below were seen in all samples of a given genotype, were observed to exhibit little variability, and were eliminated if PAR-1 fusion protein was added to the antibody binding reaction.

Antibody staining is first detected at the bend of the reflexed hermaphrodite gonad. The section of the gonad distal to the bend contains mitotic and meiotic nuclei in a syncytium. Individual meiotic nuclei at the bend are enclosed by cell membranes forming oocytes. Staining is seen associated with the cell membranes of the newly forming oocytes (Figures 5A and 5B). Since we have not determined whether this staining is at the cortex or in the membrane, we will refer to this pattern as peripheral staining. Although we see staining around the entire cell periph-

ery, only the strong staining where oocytes abut is visible in Figure 5. As the oocytes mature, the staining at the periphery becomes fainter. Peripheral staining of oocytes is not detected in *par-1* mutants (Figure 5C), and thus it is due to the PAR-1 protein.

PAR-1 Protein Is Distributed Unequally before the First Asymmetric Cleavage

Localized antibody staining is not detectable in unfertilized mature oocytes, nor in newly fertilized zygotes that are still undergoing meiotic divisions. We start to detect staining at the posterior periphery of the zygote, when both pronuclei are decondensed and the female pronucleus is just starting to migrate toward the posterior. This very early staining is faint and extends from the posterior pole to ~50% egg length (data not shown). As the cell cycle proceeds toward metaphase, the posterior peripheral staining gets stronger and becomes restricted to the posterior third of the embryo (Figure 5D). The staining becomes fainter when the first cell cycle passes metaphase and approaches telophase (data not shown).

In addition to the posterior peripheral signal, staining is also found in the internal cytoplasm of the blastomeres (Figures 5D and 5E). Using the same conditions as for wild type, we stained *par-1* mutant embryos. In nine *par-1* alleles examined, the posterior peripheral staining is absent (Figures 5F and 5G). Thus, this posterior peripheral staining is due to the PAR-1 protein. In allele *lw39*, which contains a deletion of ~5 kb in the *par-1* gene and lacks detectable PAR-1 protein (see Figure 4), the cytoplasmic staining is reduced, suggesting some staining in wild-type cytoplasm is due to unlocalized PAR-1 protein.

PAR-1 Protein Distribution Is Asymmetric in Cells of the Germline Lineage

As described above, PAR-1 is asymmetrically localized to the posterior periphery of the zygote. In addition, we observe PAR-1 staining in the cells of the germline lineage. PAR-1 protein is faintly detectable around the periphery of the newly formed P1 cell (data not shown). As the P1 cell cycle progresses, PAR-1 protein becomes localized to the posterior periphery of P1 (Figure 5E), and this localized PAR-1 is segregated into P2 at the division. Because this pattern of localization is similar to the localization of germline-specific P granules (Strome and Wood, 1983), we examined PAR-1 and P granule localization in the same embryos. As shown in Figure 6, we observe a striking correlation between PAR-1 and P granule localization. In late stage 1-cell and 2-cell embryos, both PAR-1 and P granules are localized to the posterior, and the anterior extent of the peripheral PAR-1 distribution is correlated with the anterior extent of the P granule distribution (Figures 6A and 6B).

The asymmetric PAR-1 localization at the periphery is again detected in the P2 and P3 cells. In P2 and P3, the constraints of the eggshell combine with a polarity reversal (Schierenberg, 1987) to position the two divisions along the dorsal–ventral rather than the anterior–posterior axis, with the P granules being segregated to the ventral side. In these two cells, PAR-1 becomes localized to the ventral

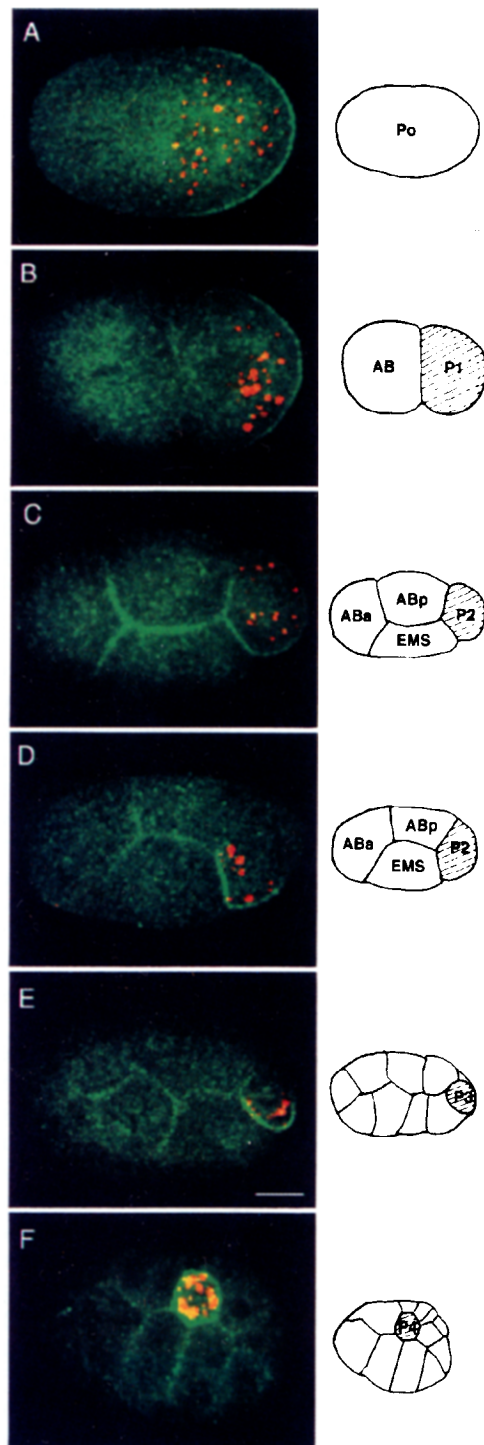


Figure 6. Correlation between PAR-1 Peripheral Distribution and P Granule Localization

(Left column) Confocal images showing PAR-1 (green) and P granule (red) distribution in wild-type embryos at different stages. Posterior is to the right.

(A–B) PAR-1 is localized at the posterior periphery in 1- and 2-cell embryos. The peripheral PAR-1 staining is correlated with the extent of P granule distribution.

(C) An early 4-cell embryo (ABa and ABp at prophase; EMS and P2 at interphase) showing the distribution of PAR-1 not only at the periphery of P2 (the posterior cell), but also at the boundaries between all four cells. EMS is ventral.

periphery (Figures 6D and 6E). Again, at the late stages of the P2 and P3 cell cycle, the P granule distribution correlates with the PAR-1 localization. Interestingly, in P4, which divides symmetrically, PAR-1 is always found to be around the entire periphery (Figure 6F). As a result, PAR-1, like P granules, is found in both daughter cells, Z2 and Z3. PAR-1 is detected around the entire periphery of both cells (data not shown). This PAR-1 staining gradually becomes fainter and is no longer detectable by the onset of morphogenesis.

Some PAR-1 staining is seen in cells outside the germline lineage. In the newly formed 4-cell embryo, PAR-1 is faintly detectable only around the periphery of P2 (data not shown). As the cell cycle progresses, PAR-1 is not only found at the periphery of P2, but is also found along the boundaries between the three somatic blastomeres (Figure 6C). By the late 4-cell stage, PAR-1 is barely detectable in the somatic cells, and the protein is mainly concentrated at the periphery on the ventral half of the P2 cell (Figure 6D). Although all of the PAR-1 phenotypes can be explained by defects in intracellular localization, the transient appearance of strong PAR-1 staining along the boundaries between cells at the 4-cell stage could reflect an additional role for PAR-1 in the cell–cell interactions that are believed to be occurring at this time (Priess and Thomson, 1987; Goldstein, 1992, 1993; Schnabel, 1994; Mello et al., 1994). Some weak and variable PAR-1 staining is also seen along the boundaries between some somatic cells at later embryonic stages (Figures 6E and 6F).

PAR-1 Kinase Activity Is Required for the Localization of P Granules and SKN-1

Two *par-1* alleles, *it51* and *it90*, carry mutations in the kinase domain. We examined PAR-1, P granule, and SKN-1 localization in these mutants. As shown in Figure 7A, the peripheral localization of PAR-1 in the *it51* zygote is indistinguishable from that of wild type (compare with Figures 5D and 6A). In 2- and 4-cell embryos, peripheral staining at the early stages of the cell cycle is similar to wild type (compare Figures 7C and 6C), but the asymmetric localization to the posterior in P1 and ventral periphery in P2 is incomplete (data not shown). In later stage embryos, the mutant PAR-1 protein is present at the periphery of a subset of blastomeres, but is not asymmetrically localized in those cells.

The P granule distribution in *it51* and *it90* embryos is abnormal and is indistinguishable from the distributions of P granules in other strong *par-1* alleles. P granules are present in the wild-type pattern in gonads and early zy-

(D) In the late 4-cell embryo (ABa and ABp at telophase; EMS at anaphase; P2 at prophase), PAR-1 is mainly concentrated at the periphery on the ventral half of P2. EMS is ventral.

(E) PAR-1 and P granules are asymmetrically localized to the ventral periphery of P3 (~15-cell stage). Scale bar, 8 μm.

(F) PAR-1 is detected around the periphery of P4 (~50-cell embryo), which no longer divides asymmetrically.

(Right column) Diagrams of embryos shown in the left column (reduced in size). The blastomeres of the germline lineage, P1–P4, are hatched.

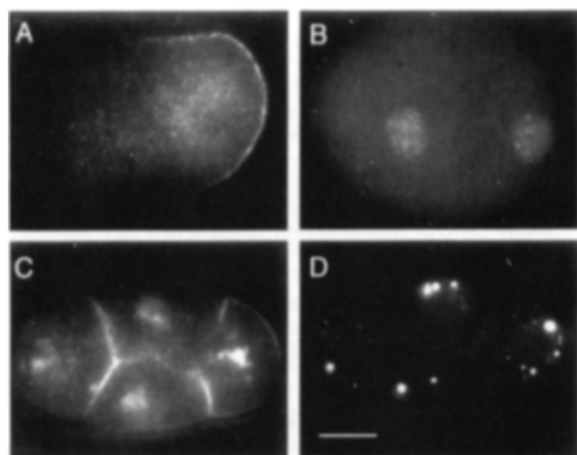


Figure 7. PAR-1, SKN-1, and P Granule Distribution in *par-1(it51)*
(A) A 1-cell *it51* embryo showing normal PAR-1 distribution.
(B) A 2-cell *it51* embryo showing SKN-1 present in equal concentration in both blastomeres, in contrast with its asymmetric localization in wild type (see Figure 1B).
(C) A 4-cell *it51* embryo showing normal PAR-1 distribution. Nuclei are stained with DAPI.
(D) The same embryo as in (C), showing abnormal P granule distribution. Scale bar, 10 μ m.

gotes before pronuclear migration (data not shown), but are undetectable in late zygotes (such as the embryo shown in Figure 7A) and in 2-cell embryos. This absence of detectable P granules in 1- and 2-cell embryos is not specific to *it51* or *it90*, but is characteristic of *par-1* mutations (unpublished data). At the 4-cell stage, P granules are easily detected, but are not localized (Figure 7D). SKN-1 protein distributions in *it51* and *it90* are also abnormal (Figure 7B). By Western blot analysis, the amount of PAR-1 protein in *it51* and *it90* is the same as wild type (data not shown).

These results suggest that the kinase activity of PAR-1 is essential for its normal function. The asymmetric localization of PAR-1 itself, at least in the zygote, does not require its own kinase activity.

Discussion

We have described the molecular cloning of *par-1*, a gene necessary for establishing asymmetry in early *C. elegans* embryos. We have shown that *par-1* encodes a conserved putative Ser/Thr kinase. The PAR-1 protein is found to be asymmetrically localized to the posterior periphery of the zygote before its mitotic division. It is then differentially distributed in the germline precursor cells, and its peripheral distribution correlates with P granule localization.

The Role of Kinase Activity in *par-1* Function

The finding of sequence similarity to Ser/Thr protein kinases suggests that PAR-1 is a protein kinase. The identification of mutations in the kinase domain in two *par-1* alleles argues strongly that the kinase activity is important for *par-1* function. Because the phenotypes caused by both alleles are indistinguishable from other strong *par-1*

mutations, we conclude that the kinase domain is essential for *par-1* function. In addition, the fact that the PAR-1 protein in these two alleles is localized properly in the zygote suggests that the kinase function is not required for PAR-1 localization.

PAR-1 and Anterior-Posterior Polarity

The initial cue that sets the anterior-posterior polarity in *C. elegans* embryos is not well understood. Mature oocytes show polarity in the acentric placement of the nucleus, but have no other obvious polarity with respect to cytoskeleton, distribution of cytoplasmic organelles, or various surface markers (Strome, 1986a). In most embryos, the polar bodies mark the future anterior and the sperm marks the future posterior. It is not clear whether the sperm entry point is restricted by preexisting oocyte polarity or whether the sperm entry point provides the cue for embryonic polarity.

Whatever the initial cue, additional events that occur during the first embryonic cell cycle appear to transduce that cue into a functional polarity in the embryo. By late prophase of the first cell cycle, the zygote is discernably polar. Cortical actin is concentrated at the anterior (Strome, 1986b), the posterior cytoplasm has acquired the ability to direct unequal cleavages (Schierenberg, 1985, 1988), and P granules are localized to the posterior (Strome and Wood, 1982, 1983). Furthermore, transient treatment of embryos with cytochalasin during the first cell cycle affects these visible polarities and alters the fates of the 2-cell stage blastomeres (Hill and Strome, 1988, 1990).

Previous work indicates that *par-1* plays a crucial role in establishing asymmetry during the first cell cycle. The *par-1* function is required for unequal first cleavage, asymmetric P granule localization, and generation of daughter cells with different fates (Kemphues et al., 1988). *par-1* is also required for the asymmetric distribution of SKN-1 (Bowerman et al., 1993).

How does the *par-1* kinase activity contribute to asymmetry in the 1-cell embryo? Although many possibilities are consistent with our results, the local accumulation of PAR-1 protein at the posterior periphery leads us to suggest that the protein acts by modifying the cortical cytoskeleton in the posterior. As a consequence of this modification, P granules and other cytoplasmic components are localized to the posterior. Modification of the posterior cortical cytoskeleton could also be required for the asymmetric placement of the spindle that leads to unequal first cleavage. In this model, many of the *par-1* phenotypes would be indirect consequences of failure in localization.

Interestingly, there may be a role for a Ser/Thr kinase in asymmetric cell division in yeast. Studies of budding yeast identified a small GTP-binding protein, Cdc42, whose activity is required for the asymmetric assembly of actin cytoskeleton at the bud site (for review see Chant, 1994). The direct downstream target of Cdc42 in yeast is unknown. However, a Ser/Thr kinase has been shown to be the probable target of the human Cdc42 homolog (Manser et al., 1994), suggesting that kinases may also be activated by yeast Cdc42.

It is possible that the role of PAR-1 protein in first cell

cycle asymmetry is not dependent upon its localization to the posterior. The protein first accumulates symmetrically at the periphery of the forming oocytes and appears to be present throughout the cytoplasm of the zygote. It could be that PAR-1 acts in an unlocalized manner, but that asymmetric localization of the protein is a mechanism to ensure a high concentration of PAR-1 in the germline lineage.

PAR-1 and Germline Cell Fate

Although it is possible that PAR-1 is only required for establishing asymmetry in the zygote, the localization of PAR-1 to the cells of the germline lineage suggests a continued role of PAR-1 in the germline. Consistent with this hypothesis, we have noted agametic phenotypes in association with insufficient activity of the maternal *par-1* gene in progeny of the following: the *par-1* homozygotes carrying *par-1*⁺ transforming DNA (YAC Y51F3), wild-type worms injected with the *par-1* antisense or sense RNA, and homozygous hermaphrodites bearing a weak allele of *par-1* (*lw7*). These observations suggest that PAR-1 function is essential for germline specification, differentiation, or maintenance. Owing to the correlation between PAR-1 distribution and P granule localization in wild-type embryos, it is possible that PAR-1 is required for the asymmetric distribution of the P granules, which, in turn, may be involved in specifying germline cell fates.

A Conserved Mechanism for Generating Intracellular Asymmetry?

Our work presented here shows that *par-1*, whose function is required for generating intracellular asymmetry in early *C. elegans* embryos, belongs to a conserved kinase subfamily. Interestingly, two members of this KIN/SNF1 kinase subfamily may also play roles in generating cellular polarity. The *Schizosaccharomyces pombe* *KIN1* gene has been shown to be required for cell growth polarity (Levin and Bishop, 1990). The human Kp78 is asymmetrically localized to the apical surface of highly polarized epithelial cells forming the lumen of the fetal pancreas, and the loss of Kp78 is associated with carcinogenesis (Parsa, 1988). The striking similarity between PAR-1 and Kp78 strongly suggests that they may have similar functions. Therefore, the study of PAR-1 function may help us understand the role of Kp78 and other kinases in establishing intracellular asymmetry.

Experimental Procedures

Strains and Alleles

The Bristol strain N2 was used as the standard wild-type strain. The genetic markers and balancer chromosomes used are the following: *LG I*, *glp-4* (*bn2*); *LG V*, *rol-4* (*sc8*), *him-5* (*e1490*), *par-1* (*b274*, *it32*, *it51*, *it60*, *it78*, *it86*, *it89*, *it90*, *it102*, *lw7*, *lw39*), *unc-76* (*e911*), *ogr-1*, *nT1 unc* (*n754*) *IV*; *V*. Nematode strains were cultured by standard techniques (Brenner, 1974).

Genetic Analysis

Alleles of *par-1* were isolated in screens for recessive, maternal-effect, embryonic lethal mutants, and the *par-1* locus was mapped 1.5 m.u. to the right of *rol-4* and 1 m.u. to the left of *unc-76* on *LG V* by three-factor recombination analysis (Kemphues et al., 1988). To facilitate molecular cloning, *par-1* was further mapped with respect to cloned markers

ogr-1 and *him-5*. The *ogr-1* locus was tagged with a transposable element Tc1 but with no observable phenotype (Telfer, 1991). A strain, *rol-4 ogr-1 unc-76/par-1*, was constructed, and standard multifactor recombination analysis was carried out, except that the presence of *ogr-1* (Tc1) was identified using the single worm PCR technique (Barstead et al., 1991). The recombinants obtained (87 between *rol-4* and *ogr-1*, 5 between *ogr-1* and *par-1*, 57 between *par-1* and *unc-76*) placed *par-1* ~0.1 m.u. to the right of *ogr-1*. Another gene, *him-5*, was also placed in this region of the physical map (S. Pennington and P. Meneely, personal communication). Similarly, *rol-4 ogr-1 par-1/him-5* was constructed, and the recombinants obtained (219 between *rol-4* and *ogr-1*, 4 between *ogr-1* and *him-5*, and 10 between *him-5* and *par-1*) indicate that the genetic distance between *him-5* and *par-1* is about three times the distance between *ogr-1* and *him-5*, placing *par-1* ~0.075 m.u. to the right of *him-5*.

Germline Transformation

Several overlapping cosmids, covering ~40 kb of genomic DNA to the right of *him-5*, were injected into the gonad of KK290 heterozygotes (*par-1* (*b274*) *unc-76/nT1 unc* (*n754*) *IV*; *V*) individually or in combination using the procedures of Mello et al. (1991). None of the cosmids rescued the *par-1* phenotype. Total genomic DNA from a yeast strain containing the YAC Y51F3 was embedded in agarose plugs and separated by pulsed field gel electrophoresis (PFGE). The position of the YAC was determined by Southern blot analysis. The YAC DNA band was excised from the gel and treated with gelase (Epicentre Technologies). The treated DNA sample was dialyzed overnight to remove free oligosaccharides and concentrated using centricon (Amicon). The concentrated DNA was mixed with plasmid DNA carrying the dominant *rol-6* marker and injected into gonads of KK290 heterozygotes. *Rol* non-*Unc-76* progeny were picked as individuals to establish transgenic lines. One line segregated *Rol Unc-76* hermaphrodites that gave viable but agametic progeny (19 of 189 *Rol Unc-76* mothers produced 94 viable progeny). Non-*Rol Unc-76* siblings were *Par* (no viable progeny).

Molecular Analyses and cDNA Cloning

Standard methods (Sambrook et al., 1989) were used for all molecular analyses except where indicated. The band corresponding to Y51F3 was excised from the gel after PFGE as described above and used in agarose-embedded restriction digestion with *AscI* (New England Biolabs) overnight at 37°C. The digested DNA was then separated by PFGE, and two fragments, 100 kb and 150 kb, were obtained. The 100 kb fragment was found by Southern blot analysis to contain *him-5* but not *ogr-1*. The 100 kb fragment was purified by GeneClean (BIO 101) and used as a probe in standard Northern blot analysis. A maternally enriched transcript of ~4.4 kb was detected. cDNAs corresponding to this transcript were obtained by screening a *C. elegans* λ ZAP cDNA library (Barstead and Waterston, 1989) using the 100 kb fragment as the probe. Since the YAC vector contained within the 100 kb fragment cross-hybridizes to the λ ZAP vector, extensive washes were carried out. Five strongly hybridizing clones were isolated, four of which hybridized to a 4.4 kb transcript by Northern blot analysis. A cDNA clone with insert size of 2.5 kb was used as the probe to further isolate a 4 kb cDNA clone, named ZC22, which also hybridized to the 4.4 kb transcript.

ZC22 was used as the probe in Southern blot analysis to detect allele-associated restriction fragment length polymorphisms. Genomic DNA was prepared from wild-type (N2) and *par-1* strains, and standard Southern blot analysis was performed.

Antisense Phenocopy Analysis

The ZC22 clone was used as a template to synthesize in vitro transcribed, capped antisense RNA. The plasmid DNA was linearized with appropriate restriction endonucleases and then treated with proteinase K, followed by phenol-chloroform extractions and ethanol precipitation. Reagents used were treated with diethyl pyrocarbonate (DEPC) to inactivate RNases. RNAs were synthesized using Stratagene in vitro transcription kit, except that a cap analog (Stratagene) and the human placental RNase inhibitor (Bethesda Research Laboratories) were added. The concentration of injected RNA ranged from 1 mg/ml up to 5 mg/ml; the higher concentrations appeared to give better results. Wild-type animals were injected with RNAs in the distal arm

of one or both gonads. Injected worms were plated out individually at 20°C to allow egg laying and were transferred to a new plate every day. Embryonic lethality (failure to hatch) was checked a minimum of 24 hr after each transfer. The early embryonic phenotypes were observed by cutting worms open in distilled H₂O and transferring the live early embryos to agar pads for microscopy.

Sequencing and Identification of the 5' Ends

DNA sequence was determined using the dideoxy chain terminator method (Sanger et al., 1977). Sequencing was performed by using Sequenase (United States Biochemicals) and double-stranded plasmid DNA; some sequences were determined by the Cornell University Biotechnology Program Automatic Sequencing Facility. Both strands of the cDNA were sequenced. The 5' end of the *par-1* gene was determined using RACE-PCR, according to the instructions of the manufacturer (Clontech). Sequence analysis was done using the FASTA (Pearson and Lipman, 1988) and BLAST programs (Altschul et al., 1990) within the Genetics Computer Group package (Madison, Wisconsin).

For sequencing of the kinase domain in the *par-1* mutant alleles, synthetic oligonucleotide primers were made that would enable the entire kinase domain to be amplified from the mutants by reverse transcription-PCR (RT-PCR). Total RNA was extracted from ~50 homozygous *par-1* hermaphrodites (N2 was used as a control) and used for reverse transcription (first-strand cDNA synthesis kit; Pharmacia) and PCR. Alterations reported in the mutant alleles (*it51* and *it90*) were sequenced from at least five independent PCR products.

Antibodies and Immunostaining

To generate polyclonal antibodies against PAR-1, we cloned a 0.65 kb fragment 3' to the kinase domain coding region into pGEX-3X (Amrad Corporation), and a fusion protein of the expected size (~45 kDa) was obtained. This protein was purified by affinity binding to agarose-immobilized glutathione and injected into two rabbits.

For affinity purification, the crude antisera were first passed over a column containing the GST protein coupled to Affigel 102 (Bio-Rad) to eliminate any anti-GST antibodies and then were affinity purified by binding to the GST-PAR-1 fusion protein coupled to the same matrix. After washing, the bound antibodies were eluted with low pH glycine and dialyzed against phosphate-buffered saline (PBS).

The ability of the purified antibody to recognize the PAR-1 fusion protein and endogenous PAR-1 protein was verified on Western blots. To detect endogenous PAR-1, we hand picked ~100 gravid hermaphrodites of each genotype. After washes with M9 buffer, the worms were boiled and ground in SDS sample buffer. The whole worm extracts were resolved by SDS-PAGE and blotted onto nitrocellulose membranes. The Western blots were performed with purified PAR-1 antibody and horseradish peroxidase-linked donkey anti-rabbit IgG. The ECL detection system (Amersham) was used.

Embryos were stained by a modification of the protocol described previously (Albertson, 1984). For each genotype examined, several samples of 10–20 hermaphrodites each were transferred to distilled H₂O (~8 µl) on polylysine slides. A coverslip (18 mm × 18 mm) was added, pressure was applied to extrude embryos, and slides were placed in liquid nitrogen for 10 min. After removal of the coverslip, slides were incubated for 15 min in -20°C absolute methanol and then transferred to room-temperature PBS. The samples were blocked in goat serum for 1 hr and incubated with PAR-1 antibody (diluted in 1% bovine serum albumin, 10% goat serum in PBS) for 1 hr at room temperature or were coincubated with anti-P granule antibody (OIC1D4 or K76 or both) at 16°C for 4 hr. Slides were washed twice for 10 min in PBS and once for 10 min in PBST (0.5% Tween in PBS) in between the two PBS washes. Slides were then incubated in secondary antibodies (for PAR-1, goat anti-rabbit FITC; for P granules, goat anti-mouse rhodamine) for 1 hr at room temperature, followed by washes as described above, except that DAPI was included in the first wash. After the final wash, slides were mounted with mounting media and viewed under fluorescence microscopes or confocal microscopes (Cornell University Biotechnology Program Confocal Microscopy Facility). Typically, each slide had between 10 and 40 embryos of each early stage (from zygote to 32-cell stage, lower numbers of later stages); a minimum of 50 slides were examined for each of N2, *it50*, *it90*, and *lw39* genotypes.

For immunofluorescence detection of PAR-1 in the gonad, a protocol adapted from the laboratory of T. Schedl (Washington University School of Medicine, St. Louis, Missouri) was used: heads of hermaphrodites were cut off (using a syringe needle) in PBS containing 0.2 mM levamisole to allow the extrusion of gonads; the worm carcasses were transferred to polylysine slides after fixation in -20°C methanol for 5 min. The same staining procedures used for embryos were applied thereafter. For SKN-1 staining, the monoclonal antibody FA2 and the procedure described by Bowerman et al. (1993) were used.

To test for specificity in the immunofluorescence assay, we incubated samples with affinity-purified PAR-1 antibody as described above except that the dilution buffer contained 4 mg/ml of the GST-PAR-1 fusion protein.

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